# Automatic Escalator

Artificial Intelligence

5/5

**Introduction**

Escalators can be designed to operate only when passengers are present and either stop or slow down when no activity has been detected for a period of time. Once a passenger has been detected, the system either starts up or speeds up back to running speed. Since the operation of the escalator is determined by the presence or absence of passengers, this kind of escalator systems are known as "service on demand" (SOD) escalators

Sensor is installed at the entry and exit of the escalator to detect the presence of passenger. When an approaching passenger is detected, the escalator will start running and complete the travelling cycle. The escalator will stop after a period of time when no further passenger is detected.

If the escalator is set going up direction, when the sensor detected a passenger coming the escalator starts to move upwards. The escalator calculate and recalculate every passenger passing by or trickle the sensor at the entry and exit sensor till the last passenger and if no further demand the escalator will stop about 80 second after that.

## PEAS description of Automatic escalator

| Agent Type | Performance Measure | Environment | Actuators | Sensors |
|---|---|---|---|---|
| Automatic escalator | fast, comfortable, safety, lower noise, save energy | shopping complexes, office buildings, private buildings, passengers | illumination for the passengers, faults display, running directions indication | Automatic start/stop travel device |

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**Performance measure:**

- Fast: escalators have the capacity to move large numbers of people.
- Comfortable: accurate stop and sooth running made passenger more comfort.
- Safety: top material and nice workmanship offer you safety escalators.
- Lower noise: escalators are quite running.
- Save energy: escalators can be designed to operate only when passengers are present and either stop or slow down when no activity has been detected for a period of time.

**Environment:**

- Shopping complexes
- Office buildings
- Private buildings
- Passengers

**Actuators:**

- Illumination for the passengers: Green light softly beams out from the space between the engaged teeth of the two neighboring steps to remind the passengers to be aware of the level sections both at the entrance and exit, resulting in increased security for the passengers.
- Faults display: When there is something wrong, while analyzing the travel condition, the faults display will show an error number to ensure an accurate and effective repair.
- Running directions indication: To direct the entrance and exit.

**Sensors:**

- Automatic start/stop travel device: A sensor near the floor plate detects the passengers approaching and leaving the escalator, hence starting and stopping the escalator automatically.

## Task environments and their characteristics

| Task Environment | Observable | Agents | Deterministic | Episodic | Static | Discrete |
|---|---|---|---|---|---|---|
| Automatic escalator | Fully | Single | Deterministic | Episodic | Static | Discrete |

**Fully-observable:** Escalators have sensors that it is already for working.

**Single Agents:** The escalators have only one agent.

**Deterministic:** The next state of the environment is completely determined by the current state and the agent's action. The escalator will work when passenger detected.

**Episodic:** The agent's experience is divided into atomic episodes and the choice of action in each episode depends only on the episode itself.

**Static:** Same environment all the time.

**Discrete:** The environment provides a fixed number of distinct percepts, actions, and environment states.